

having the user wait for the disc to wind or rewind. While tapes are still in use today as data centre backup storage, their use is gradually coming to an end. SSDs or Solid State Disks are relatively newer and a lot faster than traditional spinning hard disks.

Key points to consider for HDDs and SSDs

We are going to skip storage tapes because they no longer exist in the normal consumer landscape. The remaining two classes are HDDs and SSDs. Due to their design differences, we will address them separately.

HDDs

Hard Disk Drives or HDDs use spinning magnetic disks which store data. Typically, a hard disk has multiple disks stacked one atop another with gaps in between to allow access to the read/write heads. The speed of the hard disk depends on two factors – the speed at which the disks rotate and the time it takes for the read/write head to move from one track to another.

The figure below shows an opened up HDD. It has 3 stacked disks and the read/write heads go in the gaps between them. Normally, the read/write heads move together. But when the disk becomes fragmented, the access times spike. This happens because the heads have to move a lot more. For example, if there is a file with 100 pieces and it takes an average of 8ms to move the head, you can expect a delay of 0.8 seconds ($8\text{ms} \times 100$) when accessing those files. Also, the slower a disk spins, the slower its data transfer rates.



Internal View of HDD